Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/13/2006
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Project Title:	Gravity, Time Interactions and the Cycle of Life		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Biological physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	11706-7928	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	10/01/2000	**End Date:**	09/23/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	Note: project extended until 9/23/2007, per PI (9/11/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GA03G		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

On Earth, life exists in two basic formats, prokaryotes and eukaryotes. Eukaryotes can be further subdivided into unicellular and multicellular forms. Dictyostelium discoideum, in a sense, represents both.
This organism begins its life-cycle as a swarm of single celled amoebae feeding on bacteria. It completes it as a mature reproductive structure made up of many specialized cells with different functions. Dictyostelium's life cycle is controlled by genes and chemical messingers. In nature, this life-cycle proceeds only on a solid substrate (usually agar).

We have found that simulated altered gravity treatments perturb the organism's biochron (rate of development) significantly without causing injury, disruption or death. We have also determined that simulated altered gravity modulates construction of the erect spore-bearing stalk (sorocarp) controlling its length.

Task Description:	We are refining our flourescent cellulase measuring techniques. These techniques will provide us with the capability of quantifying the relative amounts of cellulose deposited in different D. discoideum life-cycle stages during exposure to simulated altered gravity. The objectives of this research are: 1. To determine how energy dependent obligatory cellulose synthesis is impacted by alterations in gravity. 2. To explore gravity-related phenomena (critical and probably gravity sensitive periods of development) such as changes in relative cell and stalk size and slug migration with apoptosis. 3. To study the force(s) applied to flex the cellulose stalk under conditions of altered gravity. 4. To expand overall cycle studies to include consecutive generations within the context of simulated gravity environments with a view toward performance of experiments on the Space Station. Increased knowledge of how components of the Space environment affect rates of population dispersion, differentiation, growth, maturation and reproduction of exposed biota can lead to an enhanced rate of success in human exploration, habitation and exploitation of extraterrestrial venues. Also, the use of physiological and genetic selection, as in guided evolution, will provde tools for the selection of organisms suitable for utilization on barren planetary bodies. Future NASA activities will be enhanced by the information generated by this research.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research touches on several areas of "Earth Benefits." It addresses the impacts of gravity on the rate of organismic development, apoptosis and longevity. This study also addresses the effects of gravitational forces on energy metabolism/utilization. Therefore, these studies may have human health implications. Further, Space environment factors such as exposure to unremitting, unfiltered sunlight have direct effects on the developmental program of the test subject (D. discoideum) and will provide clues to the validity of the hypothesis that the spores of an organism of this nature arriving from some extraterrestrial source, may have contributed to the bio-colonization of ancient Earth. The life cycle of D. discoideum is intimately connected with the synthesis of cellulose. This project is designed to further our knowledge of cellulose synthesis within the context of the life cycle, cellulose being a substance of major economical importance. "Benefits to America" include basic and applied research which will maintain American pre-eminence in the area of Space-related research. It may be that what is learned through this research program will impact our knowledge of the aging process and enable further extensions of the human life span. We have previously published schemes for the utilization of Dictyostelium cellulose, and believe that, others will present themselves to us, in time.
Task Progress:	Our equipment, both unique (the clinostat, centrifuge and video fluorescent morphometric microscope work station) and conjugate purification columns have been standardized. We are now amassing data. Further life-cycle studies employing clinorotation at 5.708 r.p.m. (a nominal 0.0058 g) confirmed previous results which showed that a "virtual" micro-g environment slows down the rate of completion of the life-cycle and thus, the pace of development. Further chronic centrifugation (virtual hyper-g) studies at 10 g yielded interesting results. Living in virtual hyper-g for the entire span of the life-cycle indicated that this increased g force caused a lengthening of the biochron. In other words, development proceeded more slowly. In order to determine whether simulated altered g forces have a physical impact on Dictyostelium's developmental pattern, we developed a morphometric measurement technique. Our capability, at this point, is limited to one dimension, length. The results show that of 221 sorocarps maintained at 1 g, the average length was 0.858 mm. This is a strong indication that accelerations of 10 g somehow reduce stalk length. Flourescently labelled cellulase technology employed as a tool for the quantitation of cellulose deposition during the life cycle of D. discoideum is currently under development.
Bibliography Type:	Description: (Last Updated: 04/14/2008)
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Abstracts for Journals and Proceedings	Benjaminson MA, Lehrer S, Gilchriest JA, Schonbrun L. "Gravity induced dimensional changes in developing Dictyostelium sorocarp stalks." In preparation In preparation, 2006. , Jan-2006
Abstracts for Journals and Proceedings	Benjaminson MA, Schonbrun L. "Off World Contributions to Life on Earth: Bacteria and Dictyostelium are candidates." In preparation In preparation, 2006. , Jan-2006
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Abstracts for Journals and Proceedings	Benjaminson MA, Lehrer S, Gilchriest JA, Schonbrun L, Madigan MB. "Two Special Instruments for the Study of Simulated Altered Gravity." American Society for Gravitational and Space Biology annual meeting, Arlington, VA, November 2006. Gravitational and Space Biology 2006 Nov, in press. , Nov-2006